

```
((dns-1.4.5:Network.DNS.Internal.Domain, Int]))
```

The name servers for the domain can be returned using the `lookupNS` function. For example:

```
ghci> withResolver rs $ \resolver -> lookupNS resolver
hostname
Right ["ns22.webhostfreaks.com.", "ns21.webhostfreaks.com."]
```

The type signature of the `lookupNS` function is shown below:

```
ghci> :t lookupNS
lookupNS
  :: Resolver
  -> dns-1.4.5:Network.DNS.Internal.Domain
  -> IO
      (Either
        dns-1.4.5:Network.DNS.Internal.DNSError
        [dns-1.4.5:Network.DNS.Internal.Domain])
```

You can also return the entire DNS response using the `lookupRaw` function as illustrated below:

```
ghci> :m + Network.DNS.Types

ghci> let hostname = Data.ByteString.Char8.pack "www.ubuntu.
com"

ghci> rs <- makeResolvSeed defaultResolvConf

ghci> withResolver rs $ \resolver -> lookupRaw resolver
hostname A

Right (DNSFormat
 {header = DNSHeader
 {identifier = 29504,
  flags = DNSFlags
 {qOrR = QR_Response,
  opcode = OP_STD,
  authAnswer = False,
  truncation = False,
  recDesired = True,
  recAvailable = True,
  rcode = NoErr},
  qdCount = 1,
  anCount = 1,
  nsCount = 3,
  arCount = 3},
  question = [
    Question
    {qname = "www.ubuntu.com.",
     qtype = A}],
```

```
answer = [
  ResourceRecord
  {rrname = "www.ubuntu.com.",
   rrtype = A,
   rrttl = 61,
   rdlen = 4,
   rdata = 91.189.89.103}],
  authority = [
    ResourceRecord
    {rrname = "ubuntu.com.",
     rrtype = NS,
     rrttl = 141593,
     rdlen = 16,
     rdata = ns2.canonical.com.},
    ResourceRecord
    {rrname = "ubuntu.com.",
     rrtype = NS,
     rrttl = 141593,
     rdlen = 6,
     rdata = ns1.canonical.com.},
    ResourceRecord
    {rrname = "ubuntu.com.",
     rrtype = NS,
     rrttl = 141593,
     rdlen = 6,
     rdata = ns3.canonical.com.}],
  additional = [
    ResourceRecord
    {rrname = "ns2.canonical.com.",
     rrtype = A,
     rrttl = 88683,
     rdlen = 4,
     rdata = 91.189.95.3},
    ResourceRecord
    {rrname = "ns3.canonical.com.",
     rrtype = A,
     rrttl = 88683,
     rdlen = 4,
     rdata = 91.189.91.139},
    ResourceRecord
    {rrname = "ns1.canonical.com.",
     rrtype = A,
     rrttl = 88683,
     rdlen = 4,
     rdata = 91.189.94.173}]}}
```

Please refer to the *Network.DNS hackage* Web page at <https://hackage.haskell.org/package/dns> for more information. 

By: Shakthi Kannan

The author is a free software enthusiast and blogs at shakthimaan.com